BUILTCOLAB
DIGITAL BUILT ENVIRONMENT

Como Tornar os **Edifícios Inteligentes** de Forma Sustentável, Comportável e Escalável

---

PTPC: Edifícios Energeticamente Eficientes
18/01/2023

João Pedro Dias

Software Engineering Specialist @ BUILT CoLAB
Invited Assistant Professor @ FEUP

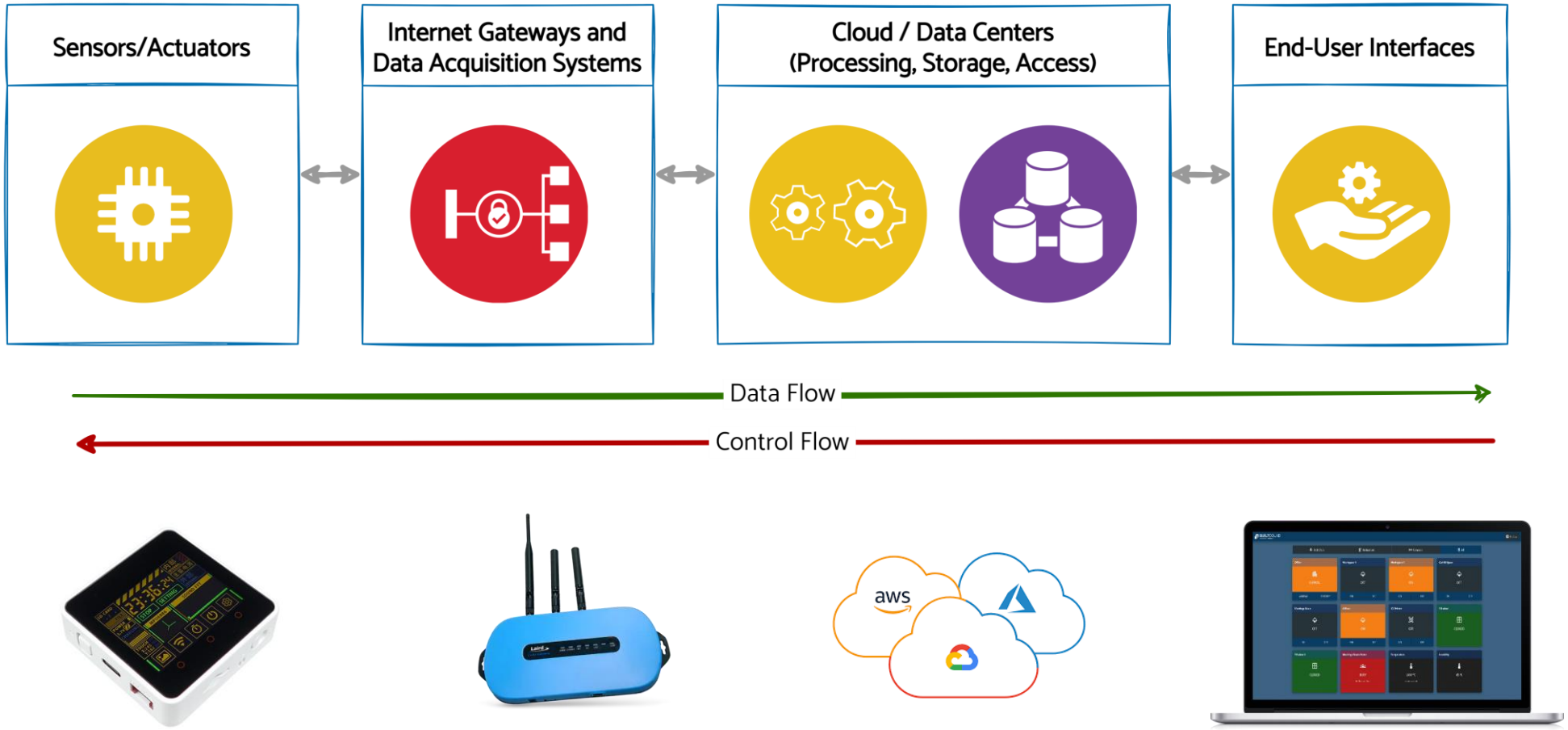
ANI AGÊNCIA NACIONAL DE INOVAÇÃO
FCT Fundação para a Ciência e a Tecnologia
Lisb@20[20]
NORTE 2020
PORTUGAL 2020
UNIÃO EUROPEIA Fundo Social Europeu

INDEX

1. Desmistificando *Buzzwords*: Internet-of-Things, Cyber-physical Systems, Smart Home, …
2. Monitorização
3. Automação
4. *Retrofit* vs. *New* (Sustentável)
5. Utilidade vs. Incômodo (Comportável)
6. Uma Questão de Escala: casas, fábricas, obras,…
7. Próximos Passos
8. Sumário

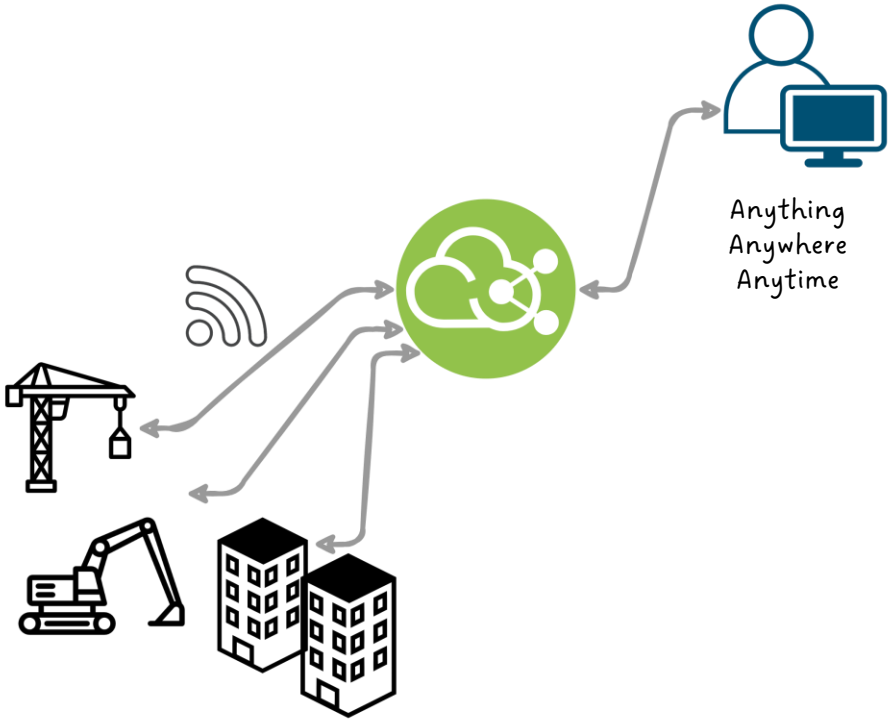
BUILTCOLAB
DIGITAL BUILT ENVIRONMENT

Desmistificando *Buzzwords*: Internet-of-Things, Cyber-physical Systems, Smart Home, …



BUILTCOLAB
DIGITAL BUILT ENVIRONMENT

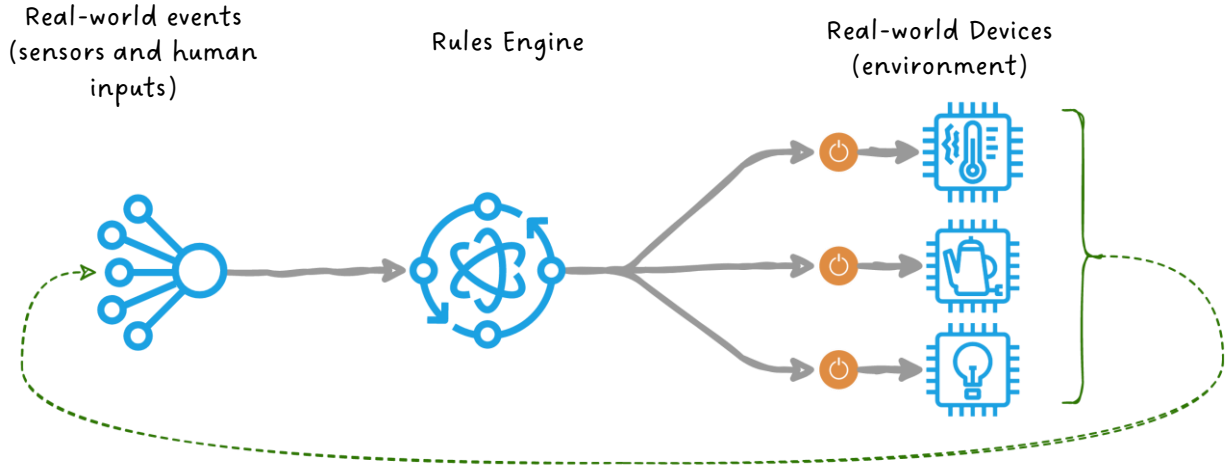
Monitorização





Automação

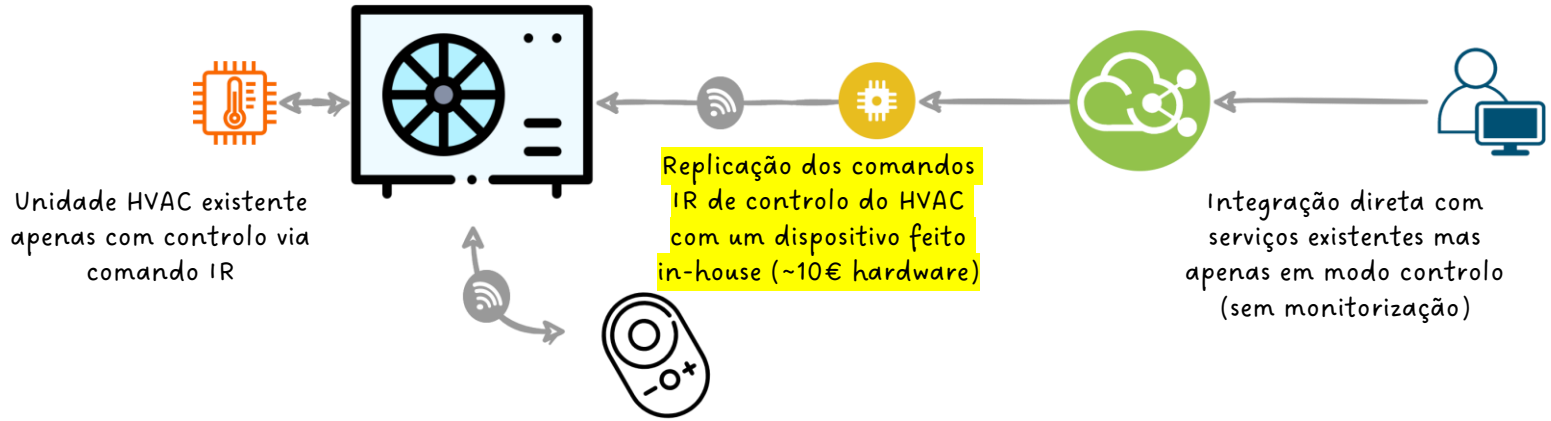


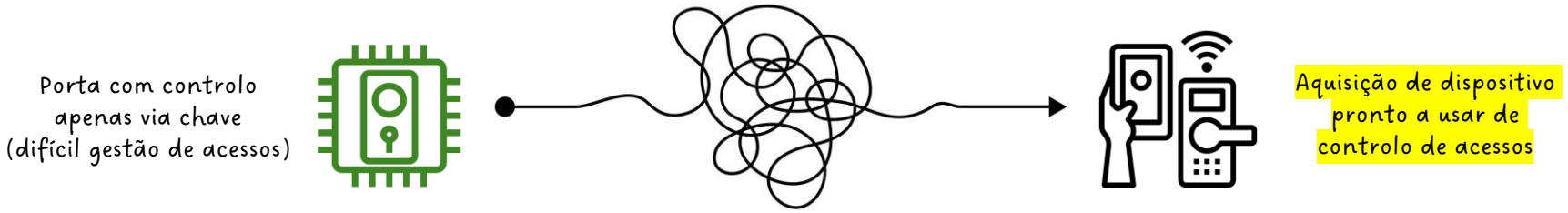


BUILTCOLAB
DIGITAL BUILT ENVIRONMENT

Retrofit vs. *New* (Sustentável)



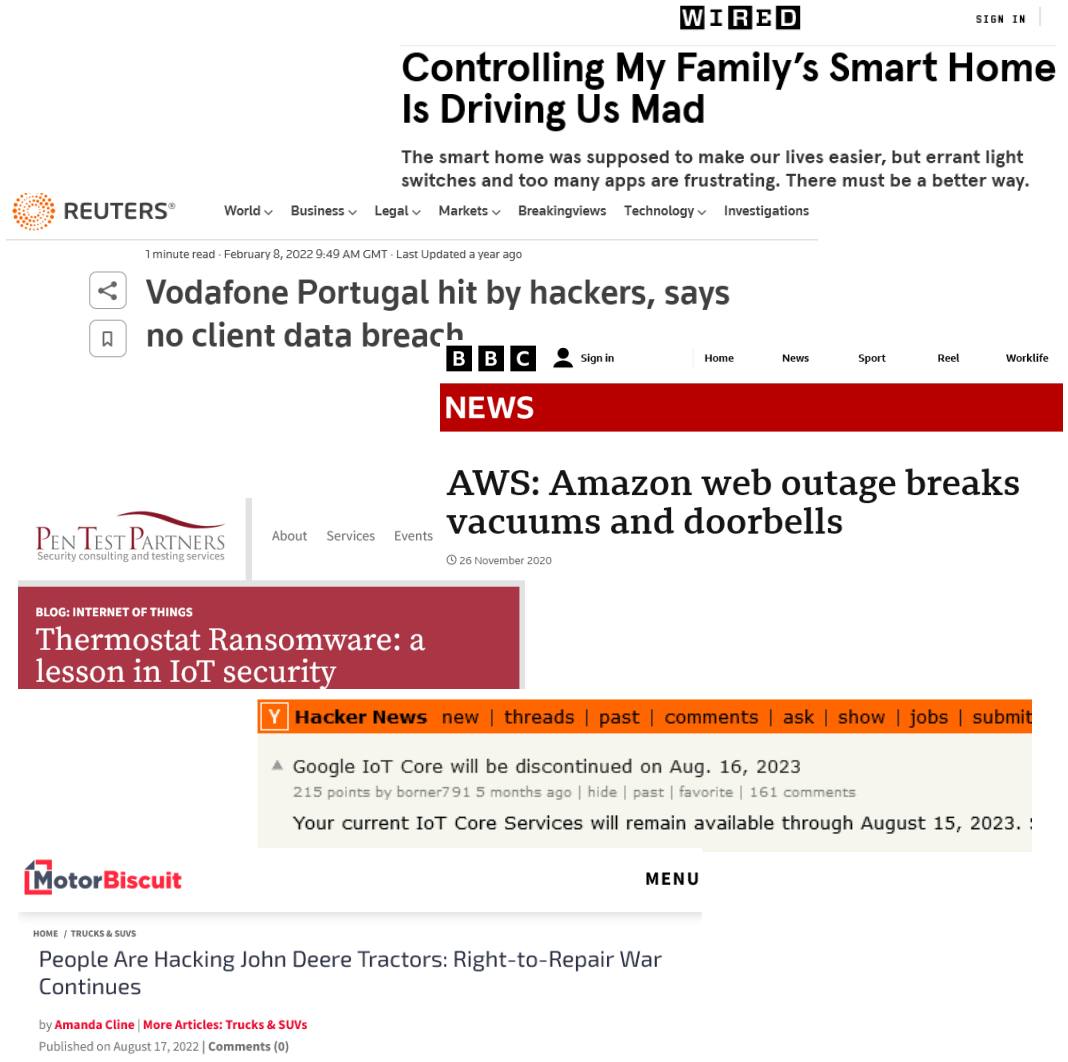




Utilidade vs. Incômodo (Comportável) —

- Controlo Remoto e Ajuste em Tempo-Real
- Monitorização e Gestão Remota
- Segurança Operacional (Alertas)
- Manutenção Preditiva
- Automatização de Tarefas
- Suporte à Decisão (data-driven)

vs.



Uma Questão de Escala: casas, fábricas, obras,…



Reasons for Proof-of-Concept Failure

Microsoft IoT Signals Report, Edition 3, October 2021

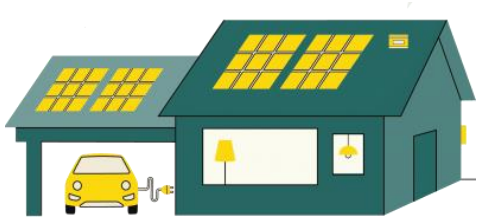
Scaling	High cost of scaling	32%
Technology	Lack of necessary technology	26%
Business	Pilots demonstrate unclear business value/ROI	25%
Complexity	Too many platforms to test	24%
Complexity	Pilot takes too long to deploy	23%
Scaling	Lack of resources/knowledge to scale	23%
Business	Hard to justify business case w/o short-term impact	23%
Complexity	Too many use cases to prove out	23%
Business	Lack of leadership support and attention	21%
Business	No clear strategy	21%
Scaling	Lack of trust in scalability platforms	20%
Business	Didn't anticipate necessary business changes	19%
Vendors	Vendors not willing to subsidize pilots	16%

Cvar N, Trilar J, Kos A, Volk M, Stojmenova Duh E. The Use of IoT Technology in Smart Cities and Smart Villages: Similarities, Differences, and Future Prospects. *Sensors*. 2020; 20(14):3897

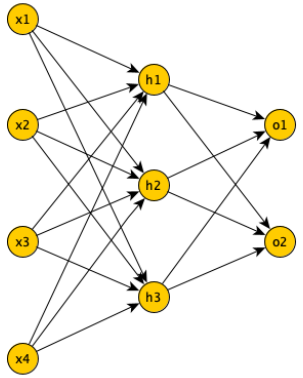
Próximos Passos



Interoperabilidade e Standards



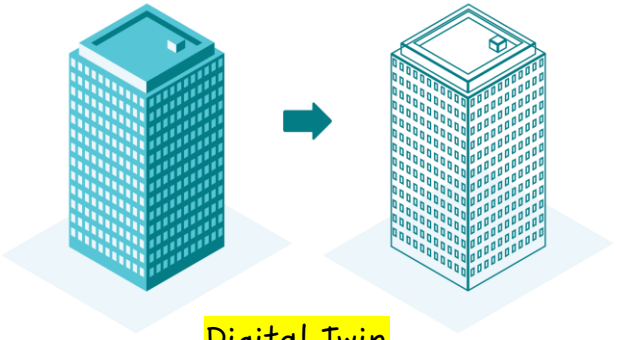
Smart Grid e Green Electricity Markets



Inteligência Artificial e Machine Learning



Gestão de Ciclo de Vida e e-Waste



Digital Twin

BUILT CoLAB

www.builtcolab.pt | info@builtcolab.pt

CONTACTOS